

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages	C++, Python, JavaScript, TypeScript, Java, SQL, Bash
Systems & Performance	Linux profiling, Spark / PySpark, Spark UI, performance tuning, multithreading, distributed data processing, batch processing
Data & Evaluation	large-scale data pipelines, batch experiments, evaluation harnesses, IR / retrieval evaluation, anti-Goodharting guardrails, reproducible benchmarking
Algorithms & ML	data structures & algorithms, combinatorial optimization (0-1 knapsack), automatic differentiation, optimization (gradient methods), PyTorch, NumPy
Cloud & DevOps	AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned backend reliability for **multi-tenant systems** (Node.js, MongoDB) serving **8+ clients** and processing **10,000+ monthly submissions**; monitored production logs across releases and integrations, investigated incidents, and shipped fixes for tenant-specific issues.
- Architected a full-stack **data annotation toolchain** (React, Node.js, MongoDB) used daily by the AI team to label and manage training data, replacing spreadsheet workflows; cut manual work by **100+ hours/month** and improved labeling throughput by **60%**.
- Remediated **20+ VAPT findings** and added integration tests to protect sensitive workflows against regressions, hardening authentication and rate limiting.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety and long-term maintainability.

Projects

Spark ETL Optimization - Large-Scale Data Pipeline

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI | 1.4B-row NYC TLC dataset

- Cut end-to-end ETL runtime **25%** (17-19 min to 14.3 min) on a **1.4B-row** dataset by replacing shuffle joins with broadcast joins (minutes to 0.05s), enabling AQE and skew-join handling, and tuning shuffle partitions, schema-aware reads, and year/month partitioning.
- Used **Spark UI** stage timings to localize shuffle and skew bottlenecks, cutting tail-latency ratio (P99/P50) from **2.14x to 1.74x**, and documented a repeatable optimization playbook.

AutoEval - Retrieval Evaluation Improvement Agent

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Built an evaluation-improvement agent that proposes new queries and labels, scores them against multiple retrieval systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and Git-backed accepted updates with report cards for an auditable, reproducible evaluation workflow.

KG2RAG-Enhanced - Multi-Hop QA Evaluation on HotpotQA

github.com/Deva-1903/KG2RAG-685-NLP

Python, sentence-transformers, networkx, Ollama | UMass CS 685 (team project)

- Built the multi-view retrieval component of a team KG-guided RAG pipeline (query decomposition, per-view retrieval, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the baseline on a **4,905-question** evaluation; the pipeline used a 0-1 knapsack token-budget formulation for evidence selection.

AgenticSearch - Provenance-First Entity Discovery Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage pipeline with typed retrieval planning, deterministic extraction before LLM fallback, output verification, and cell-level provenance; hardened with **150+ automated tests** and pipeline instrumentation for debuggability.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.